Local Alignment for Medical Vision-Language Pre-training

Huimin Yan, Xian Yang, Liang Bai, Jiye Liang *Fellow, IEEE*

***Abstract*—Establishing local semantic correspondences between medical images and their corresponding reports is crucial for effective medical vision-language pre-training. However, existing methods encounter two major challenges: (1) lesion regions in radiological images are often small, blurry, or lack clear boundaries, complicating accurate localization; and (2) medical reports typically contain redundant or non-diagnostic words, hindering precise semantic alignment. To overcome these issues, we propose MedAligner, a specialized local alignment network for medical vision-language pre-training. MedAligner employs dual encoders to extract both global and local representations and uses global contrastive learning to maintain coarse semantic consistency. To enhance local alignment, we introduce a Word-Region Alignment, which generates a learnable word-pixel similarity matrix that is sparsified to identify salient lesion regions accurately. Additionally, our Diagnostic Term Filtering dynamically samples high-importance diagnostic terms from reports, aligning them with identified lesion areas via a local contrastive loss. Importantly, we adopt a progressive training strategy that gradually refines both the input text and semantic alignment. This is achieved by reconstructing concise diagnostic reports and progressively updating word-pixel similarity, generating increasingly accurate image-text pairs. Extensive experiments demonstrate that MedAligner significantly surpasses existing approaches on tasks such as phrase grounding, image-text retrieval, and zero-shot classification, setting new benchmarks in medical vision-language pre-training.**

***Index Terms*—Local Alignment, Medical Multimodal, Vision and Language Model**

I. Introduction

MEDICAL data in real-world clinical settings is intrinsically multimodal, comprising diverse sources such as imaging, free-text reports, laboratory tests, and physiological signals. Physicians routinely integrate these heterogeneous modalities—especially radiological images and textual findings—to support accurate diagnosis and decision-making [1]. The growing availability of large-scale medical datasets has spurred interest in developing learning frameworks that can effectively leverage this multimodal information. Among these, multimodal pretraining has emerged as a powerful approach for learning transferable representations across various medical tasks [2], [3]. The visual and linguistic modalities form the core of multimodal medical data and occupy a dominant position in current multimodal pretraining research.

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Vision-language pretraining (VLP) has achieved remarkable progress in both general and medical domains. Models like Contrastive Language-Image Pretraining (CLIP) [4] have gained widespread attention due to their simple yet effective pretraining paradigm. CLIP maximizes the global representational similarity between paired images and texts through contrastive learning, while minimizing the similarity between non-paired samples. This method has been successfully applied to various medical tasks, such as disease detection, phrase grounding, and image-text retrieval [5], [6]. PubMedCLIP [7] adapts CLIP for medical data to improve medical visual question answering(VQA). BiomedCLIP [8] investigates encoder architectures and hyperparameter configurations to tailor CLIP to the medical domain.

A key challenge in medical image analysis is that diagnostic information is often subtle and localized [9]. Lesion regions are typically small with unclear boundaries, making it difficult for CLIP-style models focused on global alignment to capture fine-grained features, which reduces diagnostic accuracy [10]. To address this, fine-grained VLP models enforce local word–region consistency, improving both performance and interpretability [11]. GLoRIA [12] adds a local contrastive loss to the global alignment objective, aligning word embeddings with attention-weighted image features through bidirectional contrastive learning. This enhances task performance and yields interpretable attention maps. LRCLR [13] builds on this idea by using image encoder self-attention to identify small diagnostic regions, integrating them into textual representations via a cross-modal transformer, and applying local contrastive loss between image and text category embeddings.

Despite progress in medical vision–language pre-training, two key challenges remain. First, accurate lesion localization is difficult. Most methods use coarse region modeling, such as patch-based tokenizers (e.g., MPMA [14], GLIP [15]) or region proposal networks (e.g., Faster R-CNN [16] in Diff-MedVQA [17]). As shown in Fig. 1(a), these approaches lack the flexibility and precision needed for the ambiguous and irregular lesion appearances common in medical images. Second, redundant or non-diagnostic content in reports hinders alignment. Radiology reports often include templated phrases and background descriptions unrelated to diagnosis, which reduce text quality for image–text alignment (Fig. 1(b)). Many models (e.g., PRIOR [18], MGCA [19]) take the entire report as input, introducing noise that degrades performance. Recent methods employ large language models (LLMs), such as MedFILIP [20], to extract diagnostic entities and reduce textual complexity. However, these methods filter text unilaterally, without leveraging interactions between visual and

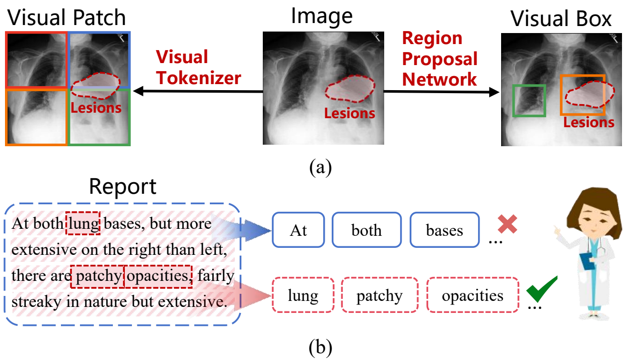

Fig. 1. Motivations behind MedAligner. (a) Lesions in radiological images are often small, blurry, or lack clear boundaries, making local region extraction challenging. Existing methods such as visual tokenization and region proposal network struggle to precisely locate such ambiguous regions. (b) Medical reports contain redundant or non-diagnostic words, hindering precise cross-modal alignment.

textual modalities. In contrast, large-scale pre-trained models like LLaVA-Med [21] and Qwen-VL [22], which leverage multimodal learning, have significantly advanced applications in image analysis and clinical decision support.

To address the aforementioned challenges, we propose MedAligner, a local alignment framework specifically designed for medical vision-language pretraining. MedAligner explicitly models local semantic correspondences between diagnostic terms and lesion areas. The framework comprises three key components: a Word-Region Alignment (WRA), a Diagnostic Term Filtering (DTF), and a Progressive Training Strategy (PTS). First, MedAligner employs dual encoders to extract both global and local representations for medical images and their corresponding reports, and enforces global semantic consistency via a contrastive objective. To achieve local alignment, WRA generates a learnable word-pixel similarity matrix, which is subsequently sparsified through two sequential sparsification layers to produce sparse word-pixel associations. This enables the identification of salient image regions corresponding to each word. DTF dynamically samples high-importance diagnostic terms from reports and aligns the sampled terms with their focus regions via a local contrastive loss, further improving the alignment quality. The proposed PTS jointly reconstructs concise diagnosis reports and refines word-pixel alignment through a progressively refinement strategy, enabling the model to learn more accurate and clinically relevant multimodal representations. To summarize, our main contributions are as follows:

- We propose MedAligner, a novel medical vision-language pretraining framework that enables precise localization of lesion regions by aligning diagnostic terms with lesion regions areas.
- We design a diagnostic term filtering to sample high-importance terms and incorporate a progressive training strategy to gradually refine cross-modal alignment, thereby enhancing the quality of multimodal representations.
- Extensive experiments show that MedAligner achieves state-of-the-art performance on tasks such as phrase grounding, image-text retrieval, and zero-shot classification tasks.

II. Related Work

A. Vision-Language Pre-training

CLIP [23] learns joint image–text representations by aligning matched pairs and contrasting unmatched pairs, achieving strong coarse-grained alignment. However, it is less effective at capturing region-level correspondences required for phrase grounding and diagnostic tasks [24], [25]. To address this, recent methods introduce fine-grained alignment mechanisms. PyramidCLIP [26] applies a semantic pyramid for intra- and cross-layer alignment, while FILIP [27] performs token-level contrastive learning through cross-modal post-interaction. FIBER [28] adopts a two-layer strategy: coarse alignment with large-scale image–text pairs, followed by fine-grained learning with bounding box annotations. GLIP [29] formulates detection as localization with pseudo-labels but depends heavily on bounding box supervision. Align and Tell [30] improves text–video retrieval by combining local contrastive learning with word-level supervision via dual transformer decoders and a captioning head. CLOC [31] extends CLIP with region–text contrastive learning by deriving region-level embeddings from bounding boxes. These advancements underscore the increasing focus on fine-grained alignment in vision-language tasks.

B. Medical Image and Report Pre-training

Recent works explore joint vision–language pre-training for medical applications [32], where fine-grained alignment is essential for linking lesion regions with textual descriptions to support clinical decisions. MPMA [14] combines global contrastive learning with patch–token alignment but suffers from noisy matches due to fixed patch sizes. GLoRIA and MGCA [33] apply token-level alignment with cross-attention but are often dominated by irrelevant regions or tokens. MedCLIP [34] aligns at the disease level using similarity matrices, missing subtle lesion–text relations. CARZero [35] improves alignment with cross-attention over image–report features, capturing complex semantic relationships. BioViL [36] and REFERS [37] integrate global contrastive and region-aware objectives but lack iterative refinement for local alignment. AFLoc [38] introduces multi-level semantic contrastive learning for annotation-free pathology localization. Huang et al. [39] propose iterative vision–language representation learning with semantic knowledge–driven report refinement to progressively extract fine-grained information. LoVT [40] performs sentence-level sub-region alignment, while PRIOR extends it with conditional reconstruction to enhance multi-scale alignment, though it still overlooks low-level features such as lesion morphology. Recent multimodal large models further advance diagnosis and lesion localization by integrating MRI, CT, and clinical reports. LLaVA-Med [21], trained on self-generated instructions, enhances medical image interpretation with conversational capability. Qwen-VL [22] supports captioning, visual question answering, and image–text matching, demonstrating strong cross-modal interaction.

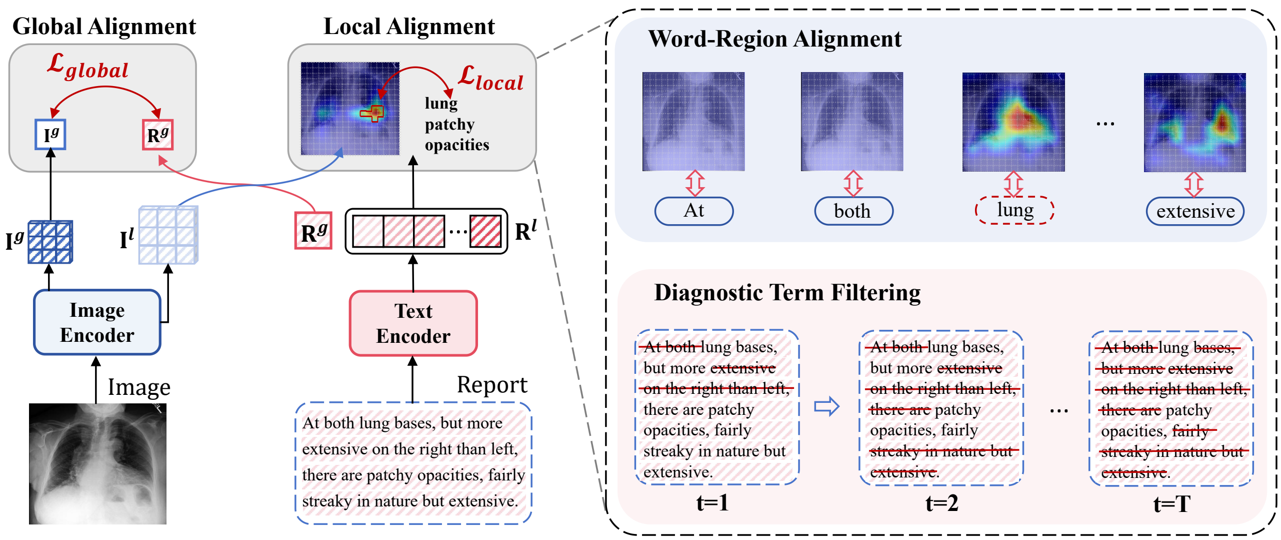

Fig. 2. The overall architecture of MedAligner comprises two key components: global alignment and local alignment. In the local alignment, MedAligner incorporates a Word-Region Alignment and a Diagnostic Term Filtering to jointly model the correspondence between pathological regions in medical images and critical semantic words in radiology reports, thereby enhancing fine-grained cross-modal alignment.

III. Method

A. Preliminary

CLIP is a powerful framework designed to align image and text by projecting both modalities into a shared latent space. This allows for the extraction of semantic relationships between medical images and their corresponding reports. The training dataset is defined as $\mathcal{D}_{\text{train}} = \{(\mathbf{I}_n, \mathbf{R}_n) \mid n = 1, 2, \ldots, N\}$, where $\mathbf{I}_n \in \mathbb{R}^{H \times W \times C}$ represents the n-th medical image with height H, width W, and C channels, while $\mathbf{R}_n \in \mathbb{R}^L$ denotes its corresponding textual report containing L words. For brevity, we omit the sample index n in subsequent equations, referring to $\mathbf{I}_n$ as simply $\mathbf{I}$ and $\mathbf{R}_n$ as simply $\mathbf{R}$.

To extract informative features from both modalities, we employ an image encoder E_{image} (e.g., ResNet [41]) and a text encoder E_{text} (e.g., BioClinical-BERT [42]). These encoders generate both global and local feature representations:

$$E_{\text{image}} : \mathbf{I} \rightarrow (\mathbf{I}^g, \mathbf{I}^l), \quad E_{\text{text}} : \mathbf{R} \rightarrow (\mathbf{R}^g, \mathbf{R}^l), \tag{1}$$

where $\mathbf{I}^g \in \mathbb{R}^D$ and $\mathbf{R}^g \in \mathbb{R}^D$ are global feature vectors that capture high-level semantic representations of the entire image and text, respectively. $\mathbf{I}^l \in \mathbb{R}^{HW \times D}$ and $\mathbf{R}^l \in \mathbb{R}^{L \times D}$ are local features that encode fine-grained details. The embedding dimension D ensures compatibility between the extracted image and text features in the latent space.

The core objective of CLIP lies in a global alignment mechanism that effectively maps image and text features into a shared latent space. This is achieved through contrastive learning, where a contrastive loss function is used to encourage high similarity between the embeddings of matching image-text pairs and low similarity between non-matching pairs. The loss for aligning images to text is defined as:

$$\mathcal{L}_g^{R \leftarrow I} = -\frac{1}{B} \sum_{n=1}^{B} \log \frac{\exp(\mathbf{I}_n^g \cdot \mathbf{R}_n^g / \tau_1)}{\sum_{k=1}^{B} \exp(\mathbf{I}_n^g \cdot \mathbf{R}_k^g / \tau_1)}, \tag{2}$$

where B is the batch size and τ_1 is a temperature parameter that controls the sharpness of the similarity distribution. The dot operator $\cdot$ refers to the inner product of vectors. Similarly, the loss for aligning text to images, $\mathcal{L}_g^{I \leftarrow R}$, is defined symmetrically to $\mathcal{L}_g^{R \leftarrow I}$. The total global alignment loss is the sum of these two components:

$$\mathcal{L}_g = \mathcal{L}_g^{R \leftarrow I} + \mathcal{L}_g^{I \leftarrow R}. \tag{3}$$

CLIP's global alignment mechanism provides a foundational approach for joint image-text representation learning. However, medical image-text tasks, such as phrase grounding require capturing fine-grained semantic correspondences.

B. Overview of the MedAligner Framework

The overall architecture is illustrated in Fig. 2. Given a medical image $\mathbf{I}$ and its corresponding radiology report $\mathbf{R}$, MedAligner first employs two independent encoders to extract both global and local representations for the image and text modalities (see Eq. 1). On this basis, we adopt the standard global contrastive learning objective used in the CLIP framework (detailed in Sec. III-A) to ensure global semantic alignment across modalities. To enhance local alignment, MedAligner introduces WRA for more accurate cross-modal local alignment. This mechanism generates a learnable word-pixel similarity matrix and sparsifies it through two sequential sparsification layers, resulting in sparse word-pixel associations, which infer the word correspondence to salient regions in the image. Moreover, DTF dynamically samples high-importance diagnostic terms from the report while filtering out redundant or non-diagnostic content. Combined with the local contrastive learning objective, this mechanism is able to enhance the semantic consistency between key diagnostic terms and corresponding salient regions. In addition, based on the sampled diagnostic terms, PTS reconstructs more focused radiology reports and progressively pairs them with the original images to generate multiple semantically aligned

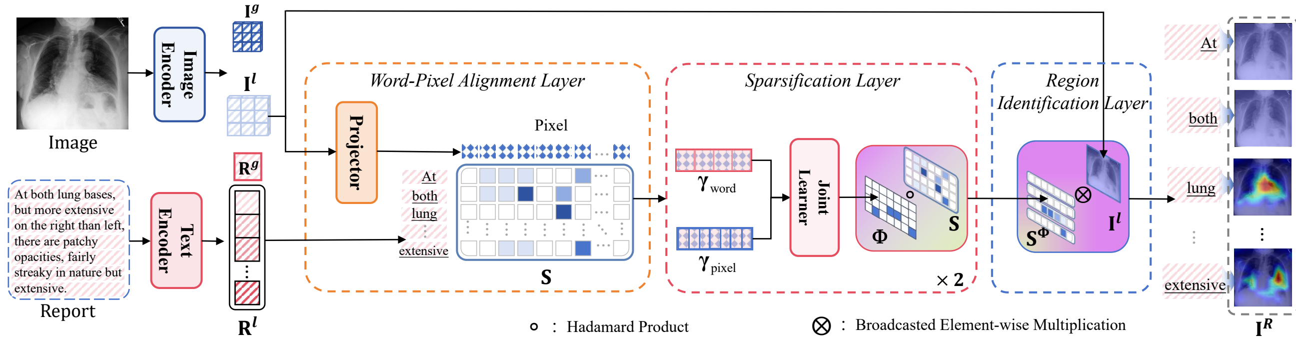

Fig. 3. Overview of the MedAligner architecture. It integrates Word-Region Alignment to capture local word-pixel associations and Diagnostic Term Filtering to extract key diagnostic terms, enabling precise semantic alignment between medical images and reports.

training pairs. Through the proposed PTS, the model progressively focuses on clinically relevant diagnostic content by reconstructing concise radiology reports based on diagnostic terms and refining the word-pixel similarity matrix via local alignment. This joint optimization enhances the model's ability to establish accurate semantic correspondence between modalities, resulting in more discriminative and interpretable cross-modal representations.

C. Word-Region Alignment

To accurately model the local alignment between medical images and radiology reports, we propose a network that can capture the association between words and image regions, as illustrated in Fig. 3.

Word-Pixel Alignment Layer. We first project the local image representation $\mathbf{I}^l$ and the local text representation $\mathbf{R}^l$ into a shared latent space using a projector. Specifically, the image representation $\mathbf{I}^l$ is transformed via a learnable function $g(\cdot)$. The transformed image features and the text features $\mathbf{R}^l$ are fed into the Word-Pixel Alignment Layer. In this layer, the image feature vector is multiplied with the text feature vector to construct a similarity matrix $\mathbf{S}$, where each row represents a word and each column corresponds to a pixel. Each element in $\mathbf{S}$ quantifies the semantic relevance between a specific word and a specific image pixel, thereby enabling local word-to-pixel alignment:

$$\mathbf{S} = \mathbf{R}^l (g(\mathbf{I}^l))^\top, \tag{4}$$

where $\mathbf{R}^l \in \mathbb{R}^{L \times D}$ denotes the local textual features of the report, and $\mathbf{I}^l \in \mathbb{R}^{HW \times D}$ represents the local image features.

Sparsification Layer. To extract high-quality region features, two sequential sparsification layers are applied to the similarity matrix $\mathbf{S}$. The first layer evaluates saliency from a unimodal perspective, while the second layer integrates information from both modalities to enhance word-to-pixel associations. This approach produces more focused attention maps, thereby improving the accuracy of lesion localization. Specifically, in each layer, we first process the row and column dimensions of $\mathbf{S}$. Processing by row can identify the key pixels of each word, while processing by column can highlight the important words of each pixel. We define the word and pixel importance vectors as $\gamma_{\text{word}} \in \mathbb{R}^L$ and $\gamma_{\text{pix}} \in \mathbb{R}^{HW}$, respectively:

$$\begin{aligned} \gamma_{\text{word}}(i) &= \frac{1}{HW} \sum_{j=1}^{HW} \mathbf{S}(i,j) + \alpha \Delta_{\text{word}}(i), \\ \gamma_{\text{pix}}(j) &= \frac{1}{L} \sum_{i=1}^{L} \mathbf{S}(i,j) + \beta \Delta_{\text{pix}}(j), \end{aligned} \tag{5}$$

where $\Delta_{\text{word}}(i)$ and $\Delta_{\text{pix}}(j)$ denote the standard deviation of similarity scores for word i and pixel j, respectively. Learnable parameters α and β balance the contribution of variance.

Since meaningful word-pixel associations arise from the joint relationship between words and pixels. we introduce a joint importance matrix $\mathbf{\Phi} \in \mathbb{R}^{L \times HW}$, where each element $\mathbf{\Phi}(i,j)$ quantifies the joint importance of word i and pixel j. It is computed by broadcasting the addition of the word and pixel importance vectors:

$$\mathbf{\Phi}(i,j) = \text{softmax}_{i,j} \left(\gamma_{\text{word}}(i) + \gamma_{\text{pix}}(j) \right). \tag{6}$$

Subsequently, we use multiplication between the joint importance matrix $\mathbf{\Phi}$ and the similarity matrix $\mathbf{S}$ to obtain a sparsified matrix $\mathbf{S}^{\Phi}$, which further emphasizes the associations between important words and salient pixels:

$$\mathbf{S}^{\Phi} = \mathbf{\Phi} \circ \mathbf{S}. \tag{7}$$

where $\circ$ denotes the Hadamard (element-wise) product. This sparsification enhances cross-modal alignment by emphasizing the salient word-pixel associations.

Region Identification Layer. We feed the sparsified similarity matrix $\mathbf{S}^{\Phi}$ and the local image representation $\mathbf{I}^l$ into the Region Identifier to further infer the correspondence between each word and semantically salient regions in the image. Specifically, $\mathbf{S}^{\Phi}$ encodes the weighted similarity between each word and every pixel location, while the local image features $\mathbf{I}^l$ represent the visual embeddings of the image across spatial dimensions. We apply a broadcasting operation to use the similarities in $\mathbf{S}^{\Phi}$ as attention weights for the corresponding pixel features. This process generates a spatial attention region for each word. The final attention region for all words is denoted as $\mathbf{I}^R_n \in \mathbb{R}^{L \times H \times W \times D}$.

$$\mathbf{I}^R = \mathbf{S}^{\Phi} \otimes \mathbf{I}^l, \tag{8}$$

where $\otimes$ denotes broadcasted element-wise multiplication, allowing each word to selectively attend to spatial regions by weighting the corresponding image features.

D. Diagnostic Term Filtering

Medical reports often contain a substantial amount of non-diagnostic textual content, which may hinder the quality of image-text alignment. To address this issue, we propose sampling diagnostically relevant words and incorporating a local contrastive learning objective to enhance their semantic consistency with salient regions in the image.

Sampling Diagnostic Terms. As illustrated in Fig. 4, we utilize the sparsified word-pixel similarity matrix $\mathbf{S}^{\Phi}$ (obtained from Eq. (7)) to compute the final word importance vector $\gamma_{\text{word}}^{\text{final}}(i)$. Gumbel-Softmax sampling is then applied to this importance vector to extract the indices of the most important words. Finally, by selecting the corresponding word embeddings from the local text representation $\mathbf{R}^l \in \mathbb{R}^{L \times D}$, we construct the representation of key diagnostic terms $\hat{\mathbf{R}}^l \in \mathbb{R}^{L' \times D}$, where L' denotes the number of selected key diagnostic terms.

Local Contrastive Learning on Word-Region Pairs. To further enhance local alignment, we introduce a contrastive-based optimization approach. Specifically, high-importance diagnostic terms and their corresponding attended regions are constructed as positive pairs to strengthen the semantic consistency between the selected textual representation $\hat{\mathbf{R}}^l$ and the corresponding image region representation $\hat{\mathbf{I}}^l$. The local alignment loss, which quantifies the alignment between terms and attended regions, is formulated as:

$$\mathcal{L}_l^{I \leftarrow R} = -\frac{1}{BL'} \sum_{n=1}^{B} \sum_{i=1}^{L'} \log \frac{\exp(\hat{\mathbf{I}}_{ni}^l \cdot \hat{\mathbf{R}}_{ni}^l / \tau_2)}{\sum_{k=1}^{L^2} \exp(\hat{\mathbf{I}}_{ni}^l \cdot \mathbf{R}_{nk}^l / \tau_2)}, \quad (9)$$

where τ_2 is a temperature parameter. The denominator encompasses all possible combinations of word-region pairs, including both positive and negative samples. Given that there are L words and each has a corresponding region, the total number of similarity scores in the matrix is L^2. After selecting L' keywords as positive pairs, the remaining $L^2 - L'$ pairs contribute to the negative samples.

In Eq.(9), $\hat{\mathbf{I}}_{ni}^l$ refers to the local image feature associated with the i-th word, which is extracted from the corresponding region in $\hat{\mathbf{I}}_n^l$ (of dimension $H \times W \times D$). To obtain a compact representation, we apply average pooling over the spatial dimensions, yielding a $1 \times 1 \times D$ feature vector, followed by a linear projection to obtain a final $1 \times D$ embedding. The representations $\mathbf{R}_{nk}^l$ and $\hat{\mathbf{R}}_{ni}^l$ correspond to the embeddings of the k-th and i-th words, extracted from the rows of $\mathbf{R}_n^l$ and $\hat{\mathbf{R}}_n^l$, respectively.

Similarly, the local alignment loss for associating pixel regions with selected keywords, $\mathcal{L}_l^{R \leftarrow I}$, is computed in an analogous manner. The total local alignment loss is then formulated as:

$$\mathcal{L}_l = \mathcal{L}_l^{R \leftarrow I} + \mathcal{L}_l^{I \leftarrow R}. \quad (10)$$

The output of this local alignment process is a refined similarity matrix $\widetilde{\mathbf{S}} \in \mathbb{R}^{L' \times HW}$, constructed by extracting the rows of $\mathbf{S}$ that correspond to the selected keyword indices.

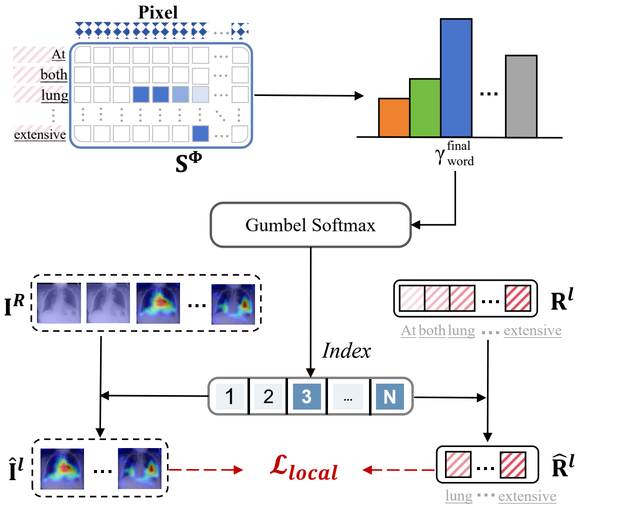

Fig. 4. Diagnostic Term Filtering. Based on the sparsified similarity matrix $\mathbf{S}^{\Phi}$, this mechanism computes word importance, samples key diagnostic terms via Gumbel-Softmax, and aligns them with attended regions using contrastive learning.

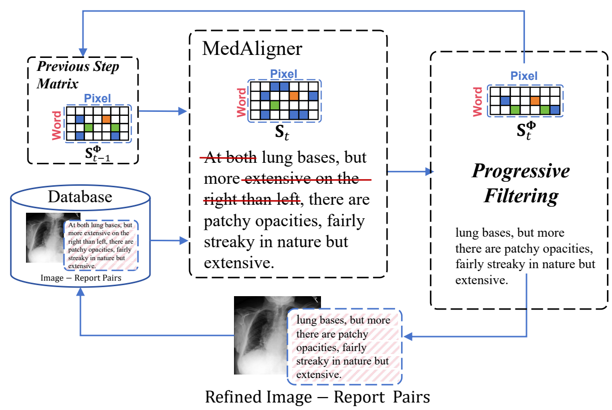

Fig. 5. The training strategy of MedAligner at step t. Important words are dynamically sampled from medical reports to reconstruct refined text, which is paired with the original image to form enhanced image-text inputs.

E. Progressive Training Strategy

Progressive Training Strategy. Building upon the previously introduced WRA and DTF, MedAligner integrates PTS, as illustrated in Fig.5, to progressively refine both textual inputs and semantic representations in a coarse-to-fine manner. At the input level, PTS reconstructs increasingly concise and focused radiology reports by sampling diagnostic terms at each step. These refined texts are progressively paired with the original medical images to form more semantically aligned and targeted training pairs. At the representation level, it refines the word-pixel similarity matrix via a smoothing mechanism, enabling more stable convergence and improved local alignment.

To model this process, we introduce a temporal index t, denoting the sparsified similarity matrix $\mathbf{S}_t^{\Phi}$ at each step, which is refined using both current and historical observations. The update rule for similarity matrix at each step t is defined as:

$$\mathbf{S}_t^{\Phi} = f\left(\delta \cdot \mathbf{S}_t + (1-\delta) \cdot \mathbf{S}_{t-1}^{\Phi}\right), \tag{11}$$

where $\mathbf{S}_{t-1}^{\Phi}$ represents the alignment state from the previous step, $\mathbf{S}_t$ is the similarity matrix computed at the current step based on Eq.(4). $\delta \in (0,1)$ is a learnable smoothing factor that balances the influence of historical and current information. $f(\cdot)$ denotes a row-removal operator in MedAligner model which aims to obtain a refined report.

Theoretical Justification in Progressive Training Strategy. To explain the effectiveness of the proposed PTS, we analyze its convergence properties.

Theorem 1 (Convergence)**.** *Let f be a row-removal operator that projects matrices to a lower-dimensional space by deleting rows. Given a smoothing factor $\delta \in (0,1)$ and bounded projected observations $\|f(\mathbf{S}_t)\|_F \leq M$ (where $\|\cdot\|_F$ denotes the Frobenius norm), the similarity matrix update rule given by Eq. (11), converges to a stable matrix:*

$$\mathbf{S}_{\infty}^{\Phi} = \sum_{k=0}^{\infty} \delta(1-\delta)^k f\left(\mathbf{S}_{t-k}\right). \tag{12}$$

Proof. Since f is a row-removal operation, it can be represented as a linear projection matrix $\mathbf{P} \in \mathbb{R}^{k \times m}$ (where each row contains exactly one "1" and others are "0"). For any matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$:

$$f(\mathbf{X}) = \mathbf{PX}.$$

Substituting into the update rule (11):

$$\mathbf{S}_{t-1}^{\Phi} = \delta \mathbf{PS}_{t-1} + (1-\delta)\mathbf{PS}_{t-2}^{\Phi}.$$

Substituting back into $\mathbf{S}_t^{\Phi}$:

$$\begin{aligned}\mathbf{S}_t^{\Phi} &= \delta \mathbf{PS}_t + (1-\delta)\left[\delta \mathbf{PS}_{t-1} + (1-\delta)\mathbf{PS}_{t-2}^{\Phi}\right] \\ &= \delta \mathbf{PS}_t + \delta(1-\delta)\mathbf{PS}_{t-1} + (1-\delta)^2\mathbf{PS}_{t-2}^{\Phi}.\end{aligned}$$

Continuing this expansion to the initial iteration $t=0$:

$$\mathbf{S}_t^{\Phi} = \sum_{k=0}^{t} \delta(1-\delta)^k \mathbf{PS}_{t-k} + (1-\delta)^{t+1}\mathbf{PS}_0^{\Phi}.$$

For the infinite series $\sum_{k=0}^{\infty} \delta(1-\delta)^k \mathbf{PS}_{t-k}$, the boundedness condition $\|\mathbf{PS}_t\|_F \leq M$ implies:

$$\begin{aligned}\left\|\sum_{k=0}^{\infty} \delta(1-\delta)^k \mathbf{PS}_{t-k}\right\|_F &\leq \sum_{k=0}^{\infty} \delta(1-\delta)^k \|\mathbf{PS}_{t-k}\|_F \\ &\leq M\sum_{k=0}^{\infty} \delta(1-\delta)^k.\end{aligned}$$

Evaluating the geometric series:

$$\sum_{k=0}^{\infty} \delta(1-\delta)^k = \delta \cdot \frac{1}{1-(1-\delta)} = 1.$$

Thus, the series converges absolutely with:

$$\left\|\sum_{k=0}^{\infty} \delta(1-\delta)^k \mathbf{PS}_{t-k}\right\|_F \leq M.$$

The norm of the initial term decays as:

$$\|(1-\delta)^{t+1}\mathbf{PS}_0^{\Phi}\|_F \leq (1-\delta)^{t+1}\|\mathbf{P}\|_F\|\mathbf{S}_0^{\Phi}\|_F.$$

Since $\|\mathbf{P}\|_F = \sqrt{k}$ (Frobenius norm of the projection matrix) and $0 < \delta < 1$, we have $(1-\delta)^{t+1} \to 0$ as $t \to \infty$.

Combining all results, as $t \to \infty$:

$$\mathbf{S}_t^{\Phi} = \underbrace{\sum_{k=0}^{t} \delta(1-\delta)^k \mathbf{PS}_{t-k}}_{\text{Converges to } \mathbf{S}_{\infty}^{\Phi}} + \underbrace{(1-\delta)^{t+1}\mathbf{PS}_0^{\Phi}}_{\text{Vanishes}} \to \mathbf{S}_{\infty}^{\Phi}.$$

Hence, the similarity matrix converges to:

$$\mathbf{S}_{\infty}^{\Phi} = \sum_{k=0}^{\infty} \delta(1-\delta)^k f(\mathbf{S}_{t-k}).$$

This completes the proof. □

Lemma 1 (Alignment Error Bound)**.** *Let $\mathbf{S}^* \in \mathbb{R}^{L \times HW}$ denote the ground-truth word-to-pixel alignment matrix. Assume that the projected similarity matrices $\mathbf{S}_t$ satisfy a bounded deviation with respect to $\mathbf{S}^*$, i.e., the error in the projected similarity matrices is bounded by ϵ:*

$$\|f(\mathbf{S}_t) - \mathbf{S}^*\|_F \leq \epsilon, \quad \forall t, \tag{13}$$

where $f(\cdot)$ is the row-removal projection operator. Then, under the progressive update rule

$$\mathbf{S}_t^{\Phi} = f\big(\delta \mathbf{S}_t + (1-\delta)\mathbf{S}_{t-1}^{\Phi}\big), \quad \delta \in (0,1), \tag{14}$$

the alignment error is bounded as

$$\|\mathbf{S}_t^{\Phi} - \mathbf{S}^*\|_F \leq (1-\delta)^t\|\mathbf{S}_0^{\Phi} - \mathbf{S}^*\|_F + \big(1-(1-\delta)^t\big)\epsilon. \tag{15}$$

In particular, as $t \to \infty$, we have

$$\limsup_{t\to\infty} \|\mathbf{S}_t^{\Phi} - \mathbf{S}^*\|_F \leq \epsilon. \tag{16}$$

Proof. From the update rule we have

$$\mathbf{S}_t^{\Phi} - \mathbf{S}^* = f\big(\delta \mathbf{S}_t + (1-\delta)\mathbf{S}_{t-1}^{\Phi}\big) - \mathbf{S}^*.$$

Using the linearity of $f(\cdot)$, this can be rewritten as

$$\mathbf{S}_t^{\Phi} - \mathbf{S}^* = \delta f(\mathbf{S}_t - \mathbf{S}^*) + (1-\delta) f(\mathbf{S}_{t-1}^{\Phi} - \mathbf{S}^*).$$

Taking Frobenius norm and applying the triangle inequality yields

$$\|\mathbf{S}_t^{\Phi} - \mathbf{S}^*\|_F \leq \delta\|f(\mathbf{S}_t - \mathbf{S}^*)\|_F + (1-\delta)\|\mathbf{S}_{t-1}^{\Phi} - \mathbf{S}^*\|_F.$$

By assumption, $\|f(\mathbf{S}_t - \mathbf{S}^*)\|_F \leq \epsilon$. Thus,

$$\|\mathbf{S}_t^{\Phi} - \mathbf{S}^*\|_F \leq \delta\epsilon + (1-\delta)\|\mathbf{S}_{t-1}^{\Phi} - \mathbf{S}^*\|_F.$$

Unrolling the recursion for t steps gives

$$\|\mathbf{S}_t^{\Phi} - \mathbf{S}^*\|_F \leq (1-\delta)^t\|\mathbf{S}_0^{\Phi} - \mathbf{S}^*\|_F + \delta\sum_{k=0}^{t-1}(1-\delta)^k\epsilon.$$

The geometric series can be evaluated as

$$\delta\sum_{k=0}^{t-1}(1-\delta)^k = 1-(1-\delta)^t.$$

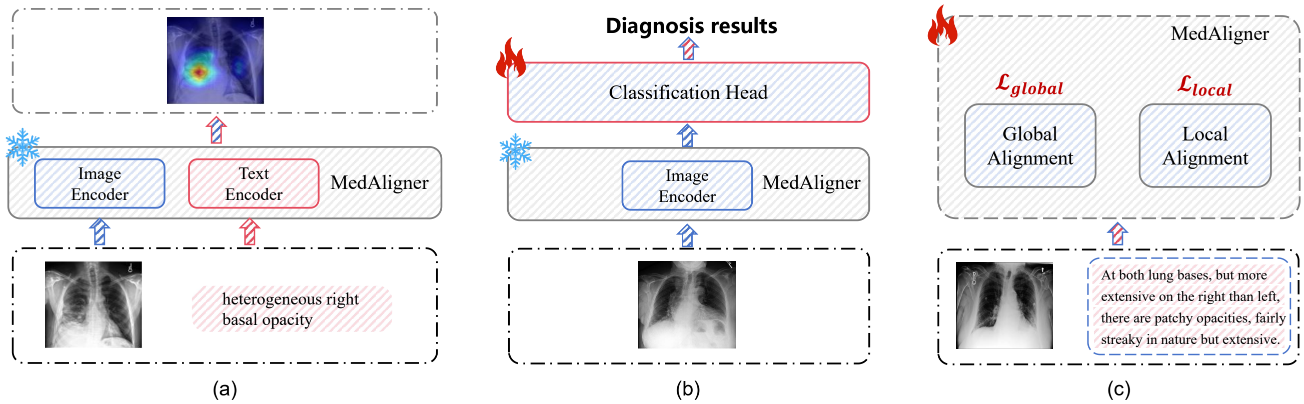

Fig. 6. Applications of the MedAligner in key medical tasks: (a) Lesion localization is achieved through progressive word-region alignment; (b) Disease diagnosis is facilitated via image-level classification; (c) Pre-training on unpaired image-text data using a local-to-global alignment strategy.

Therefore,

$$\|\mathbf{S}_t^\Phi - \mathbf{S}^*\|_F \le (1-\delta)^t \|\mathbf{S}_0^\Phi - \mathbf{S}^*\|_F + (1-(1-\delta)^t)\epsilon.$$

Finally, taking the limit as $t \to \infty$, we obtain

$$\limsup_{t\to\infty} \|\mathbf{S}_t^\Phi - \mathbf{S}^*\|_F \le \epsilon.$$

This completes the proof. □

The results jointly demonstrate that as the number of iterations increases, the alignment error monotonically decreases and eventually stabilizes within a narrow range, determined by the noise level ϵ. This indicates that progressive training is guaranteed to converge, rather than diverge, ultimately achieving a theoretically reliable alignment. Importantly, this refinement operates in tandem with the progressive report reconstruction introduced earlier-together. While the DTF gradually constructs more discriminative diagnostic descriptions, the smoothing of $\mathbf{S}$ ensures that local visual-textual alignment becomes increasingly stable and semantically meaningful. In practice, the number of progressive steps $T = 3$ is treated as a tunable hyperparameter and is selected based on empirical analysis to balance performance and computational cost, as validated in the experimental section.

The overall learning objective, $\mathcal{L}$, is formulated as the average alignment loss over all iterations:

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T} \mathcal{L}_{\text{align}}^t, \tag{17}$$

where T denotes the total number of training steps. The alignment loss at each step is given by:

$$\mathcal{L}_{\text{align}}^t = \mathcal{L}_g^t + \mathcal{L}_l^t, \tag{18}$$

where $\mathcal{L}_g^t$ and $\mathcal{L}_l^t$ are the global and local alignment losses, respectively, as defined in Eq. (3) and Eq. (10). Note that for brevity, the step index t was omitted in the previous equations. Optimizing this objective function enables the model to dynamically focus on lesion regions in medical images and the key semantic terms of diseases in medical reports.

IV. Medical Application for Local Alignment

MedAligner is designed to address the challenge of aligning medical images with corresponding report content at the local level, a critical step for accurate diagnosis and clinical decision-making. It learns fine-grained word–region correspondences and selectively attends to diagnostically relevant terms, improving semantic alignment. The following sections present its applications in Lesion Localization, Disease Diagnosis, and Base Model Pre-training with Unpaired data, highlighting its potential to optimize clinical workflows and enhance diagnostic accuracy.

A. Lesion Localization

Precise lesion localization is critical for medical image interpretation, particularly in radiology, where accurate identification of abnormal regions directly impacts diagnosis and treatment decisions. MedAligner introduces WRA that dynamically extracts high-importance diagnostic terms from radiology reports. By adopting PTS, it incrementally refines the WRA to achieve high-precision localization (Fig. 6(a)).

Given a medical image $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ and a lesion-related phrase $\mathbf{R} \in \mathbb{R}^L$, we input them into pretrained MedAligner. The image is encoded into local visual features $\mathbf{I}^l \in \mathbb{R}^{HW\times D}$, while the phrase is encoded into token embeddings $\mathbf{R}^l \in \mathbb{R}^{L\times D}$. MedAligner then produces a heatmap that highlights the image regions corresponding to the given phrase, indicating the alignment between phrase and image. This framework effectively handles complex lesions, offering precise, interpretable localization that enhances clinical confidence and decision-making.

B. Disease Diagnosis

Accurate disease diagnosis requires both localizing suspicious regions and interpreting their clinical significance. To address this, the proposed MedAligner strengthens cross-modal understanding by integrating local visual features with complex medical semantics (Fig. 6(b)).

Given a medical image $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, the pretrained MedAligner encodes it into a global image representation

TABLE I
THE USAGE OF THE DATASET EMPLOYED BY THE MODEL IN PRE-TRAINED AND MEDICAL TASKS.

Dataset	Size/Images	Annotations	Pre-training	Phrase Grounding	Object Detection	Image-Text Retrieval	Zero-shot Classification
MIMIC-CXR	377,110	Paired images and reports	✓				
MS-CXR	206	Bounding boxes, radiology phrase		✓			
MIMIC-5 $\times$ 200	1,000	Radiology text, 5 diseases				✓	✓
COVID	5,162	COVID-19/normal labels					✓
RSNA Pneumonia	12,024	Pneumonia/normal labels			✓		✓

$\mathbf{I}^g \in \mathbb{R}^D$ via Equation (1). A classification head, composed of fully connected layers, then predicts the disease category based on $\mathbf{I}^g$ as follows:

$$\hat{\mathbf{y}} = \mathrm{Softmax}(W_c \mathbf{I}^g + b_c) \tag{19}$$

where $\hat{\mathbf{y}}$ denotes the predicted probability distribution over disease classes, and W_c, b_c are learnable parameters.

MedAligner enhances the discriminative capacity of the encoder by generating a series of semantically refined image-text pairs through PTS (Sec. III-E). During training, both global and local alignment objectives are jointly optimized, while the fine-tuning stage further adapts the classification head to task-specific data, thereby improving diagnostic performance. This approach not only increases predictive accuracy but also ensures scalability and robustness for clinical deployment.

C. Pre-training of the Base Model on Unpaired Data

In medical scenarios, unpaired image-text data is prevalent, posing significant challenges for model training. To address this, MedAligner employs WRA that dynamically samples high-importance diagnostic terms from clinical reports. PTS is further introduced to incrementally refine semantic alignment, enabling effective alignment learning even in the absence of paired supervision (Fig. 6(c)).

Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ and report $\mathbf{R} \in \mathbb{R}^L$, MedAligner extracts local representations $\mathbf{I}^l \in \mathbb{R}^{HW\times D}$ and $\mathbf{R}^l \in \mathbb{R}^{L\times D}$ (Eq.(1)) to train the local alignment. After diagnostic terms sampling, the sampled terms are recomposed into sentences, whose global representation $\hat{\mathbf{R}}^g$ (via the text encoder) is aligned with the global image representation $\mathbf{I}^g$. The image-to-text alignment loss is:

$$\hat{\mathcal{L}}_g^{R\leftarrow I} = -\frac{1}{B}\sum_{n=1}^{B} \log \frac{\exp(\mathbf{I}_n^g \cdot \hat{\mathbf{R}}_n^g/\tau_2)}{\sum_{k=1}^{B}\exp(\mathbf{I}_n^g \cdot \hat{\mathbf{R}}_k^g/\tau_2)}, \tag{20}$$

where B is the batch size, τ_2 is a temperature parameter, and $\cdot$ denotes the inner product. The text-to-image loss $\hat{\mathcal{L}}_g^{I\leftarrow R}$ is defined similarly, and the global alignment loss is:

$$\hat{\mathcal{L}}_g = \hat{\mathcal{L}}_g^{R\leftarrow I} + \hat{\mathcal{L}}_g^{I\leftarrow R}. \tag{21}$$

The total loss combines local and global alignment:

$$\mathcal{L}_{unpair} = \mathcal{L}_l + \hat{\mathcal{L}}_g \tag{22}$$

This framework allows MedAligner to leverage unpaired data, reducing manual annotation costs and enhancing the applicability of vision-language models in clinical contexts.

V. EXPERIMENT

A. Experimental Settings

1) Datasets: We utilized multiple datasets for model pre-training and evaluation. Table I summarizes their annotations and tasks, including phrase grounding, object detection, image-text retrieval, and zero-shot classification, for a comprehensive assessment of MedAligner.

MIMIC-CXR: The MIMIC-CXR v2.0.0 dataset includes 377,110 chest X-rays and 227,835 reports from 65,379 patients [43]. Only the Findings and Impression sections were used as text input. Following the strategy proposed in MedCLIP [34], the data were split into a pre-training set (354,615 images, 201,063 reports) and an evaluation set (MIMIC-5 $\times$ 200), as described later.

MS-CXR: A phrase grounding dataset with radiologist-verified bounding boxes and text for eight chest diseases [44], used to align disease phrases with image regions.

MIMIC-5×200: Following GLoRIA [12], we derived a balanced multi-class subset of MIMIC-CXR for zero-shot classification and image-text retrieval, including five diseases: atelectasis, cardiomegaly, consolidation, edema, and pleural effusion.

RSNA Pneumonia: A binary chest X-ray dataset labeled as pneumonia or normal, with bounding boxes for detection [45]. Following MedCLIP [34], we sampled a balanced subset: 8,486 images for training and 3,538 for testing.

COVID: A binary chest X-ray dataset for COVID-19 detection, sampled into COVID-19 and normal classes [46]. Following MedCLIP, we used 2,162 images for training and 3,000 for testing.

NIH ChestX-ray: A dataset of 112,120 frontal chest X-rays from 30,805 patients [47], labeled with 14 thoracic disease categories: atelectasis, consolidation, infiltration, pneumothorax, edema, emphysema, fibrosis, effusion, pneumonia, pleural thickening, cardiomegaly, nodule, mass, and hernia. Model fine-tuning and evaluation followed the official train/test split.

2) Baselines: We compared several state-of-the-art medical image-text pretraining methods, including MedCLIP[34], MGCA[19], MedKLIP[48], CLEFT[49], MAVL[50], PRIOR[18], CARZero[35], and AFLoc[38]. Since MedCLIP and CLEFT were pretrained on CheXpert dataset, we retrained their models on the MIMIC-CXR dataset for fair comparison.

3) Implementation Details: Our model was pretrained on the MIMIC-CXR dataset for 10 epochs with a batch size of 48 on a single NVIDIA A100 40GB GPU, 512GB RAM, and AMD EPYC 7352 CPU. Parameter updates were performed using the Adam optimizer (weight decay: 1×10^{-6}), with an initial learning rate of 1×10^{-5}, dynamically adjusted based

TABLE II
THE PERFORMANCE OF THE PHRASE GROUNDING TASK IS EVALUATED ON THE MS-CXR DATASET USING CONTRAST-TO-NOISE RATIO (CNR) SCORES ACROSS EIGHT DISEASE CATEGORIES.

Model	Atelectasis	Cardiomegaly	Consolidation	Lung Opacity	Edema	Pneumonia	Pneumothorax	Pleural Effusion
MedCLIP	0.56±0.06	0.53±0.03	0.58±0.02	0.56±0.08	0.44±0.06	0.62±0.03	0.47±0.02	0.51±0.08
GLoRIA	0.61±0.04	0.53±0.02	0.62±0.03	0.54±0.02	0.51±0.03	0.73±0.14	0.59±0.09	0.90±0.05
BioViL	0.54±0.01	0.50±0.04	0.59±0.04	0.53±0.03	0.47±0.02	0.46±0.01	0.49±0.05	0.58±0.03
MGCA	0.63±0.05	0.57±0.02	0.65±0.05	0.56±0.08	0.52±0.06	0.83±0.15	0.52±0.02	0.82±0.13
MedKLIP	0.56±0.02	0.53±0.07	0.59±0.01	0.59±0.11	0.49±0.04	0.68±0.13	0.59±0.01	0.64±0.07
PRIOR	0.65±0.08	0.56±0.01	0.59±0.04	0.66±0.09	0.55±0.03	0.67±0.06	0.66±0.10	0.82±0.09
CLEFT	0.69±0.10	0.57±0.02	0.65±0.02	0.57±0.08	0.50±0.04	0.73±0.10	0.60±0.12	0.80±0.10
MAVL	0.58±0.08	0.54±0.01	0.55±0.05	0.64±0.08	0.51±0.02	0.71±0.09	0.60±0.07	0.76±0.13
CARZero	0.68±0.09	0.52±0.07	0.69±0.09	0.56±0.02	0.52±0.03	0.84±0.12	0.63±0.05	0.97±0.12
AFLoc	0.79±0.05	0.59±0.02	0.77±0.04	0.69±0.03	0.63±0.01	0.75±0.06	0.68±0.05	0.99±0.03
MedAligner (Our)	**0.83**±0.06	**0.61**±0.01	**0.81**±0.02	**0.72**±0.07	**0.73**±0.08	**0.85**±0.05	**0.72**±0.08	**1.02**±0.10

on the ReduceLROnPlateau strategy. The Gumbel-Softmax is applied with the hyperparameter `hard` set to `False`. During image preprocessing, images were resized to 256×256 and randomly cropped to 224×224 to enhance model robustness.

B. Experimental Results

We evaluated MedAligner through five experiments in three application scenarios: lesion localization, disease diagnosis, and unpaired image–text pretraining. For lesion localization, phrase grounding was used to align diagnostic phrases with pathological regions and visualize disease areas. For disease diagnosis, image–text retrieval assessed global semantic alignment, object detection measured lesion identification, and zero-shot classification tested generalization to unseen diseases. Finally, unpaired image–text experiments evaluated pretraining without paired supervision. Overall, these tasks provide a comprehensive assessment of MedAligner's localization, diagnostic, and representation learning capabilities.

Phrase Grounding. Phrase grounding associates textual phrases (e.g., disease descriptions or anatomical terms) with corresponding regions in medical images, offering precise diagnostic insights and enhancing model interpretability. Table II presents the phrase grounding results on the MS-CXR dataset. Using the Contrast-to-Noise Ratio (CNR) [51] as the evaluation metric, MedAligner achieved the highest CNR across eight disease categories, outperforming MGCA and PRIOR. Heatmaps generated with Grad-CAM (Figure 7) further illustrate MedAligner's ability to accurately localize lesion sites and align disease-related phrases with image regions. These findings highlight MedAligner's superior precision and interpretability in phrase grounding tasks.

Image-Text Retrieval. To assess the matching effectiveness between medical images and textual descriptions, we conducted experiments on the MIMIC-5 $\times$ 200 dataset. For each query image, cosine similarity was computed between the `[cls]` token representation and candidate sentences. Retrieval quality was measured using Precision@K, where higher values indicate stronger alignment between retrieved reports and the image's disease category. Results are shown in Table III. PRIOR and AFLoc benefit from fine-grained information; however, MedAligner achieves the best performance, confirming that WRA improves cross-modal semantic alignment and image–text matching.

TABLE III
THE IMAGE-TEXT RETRIEVAL TASK IS COMPARED WITH STATE-OF-THE-ART METHODS ON THE MIMIC-5 $\times$ 200 DATASET, AND PRECISION (%) SCORES ARE REPORTED.

Model	Prec@1	Prec@2	Prec@5	Prec@10
MedCLIP	44.40	43.45	44.50	45.58
GLoRIA	42.38	45.65	47.74	41.98
BioViL	48.81	47.88	50.05	32.88
MGCA	50.06	49.77	49.05	47.53
MedKLIP	50.00	50.00	49.42	50.00
PRIOR	53.19	51.72	51.89	41.69
CLEFT	52.75	50.31	48.75	49.81
MAVL	50.00	50.62	51.06	49.97
CARZero	50.00	50.00	51.65	47.38
AFLoc	54.37	52.78	49.79	43.20
MedAligner (Our)	**55.88**	**54.08**	**54.05**	**50.56**

Beyond alignment, the performance of MedAligner's image encoder is critical for downstream tasks, where the learned image features are applied to object detection and zero-shot classification. Object detection assesses its ability to localize lesions and distinguish fine-grained disease categories, while zero-shot classification measures its capacity to generalize to unseen categories. In combination, these tasks provide a comprehensive evaluation of the encoder's robustness and transferability.

Object Detection. The RSNA Pneumonia dataset, containing binary labels for pneumonia and normal cases, was used to evaluate the performance of image-text pretraining models in object detection tasks. We used three metrics: Intersection over Union (IoU), Dice coefficient (Dice), and Accuracy (Acc). IoU measures pixel-level overlap between predicted and ground-truth regions, Dice assesses similarity in predicted and ground-truth pixels, and Acc evaluates overall classification accuracy. Higher values across these metrics indicate better model performance.

Table IV presents the results of the object detection task. MedAligner leverages word-to-pixel alignment to establish

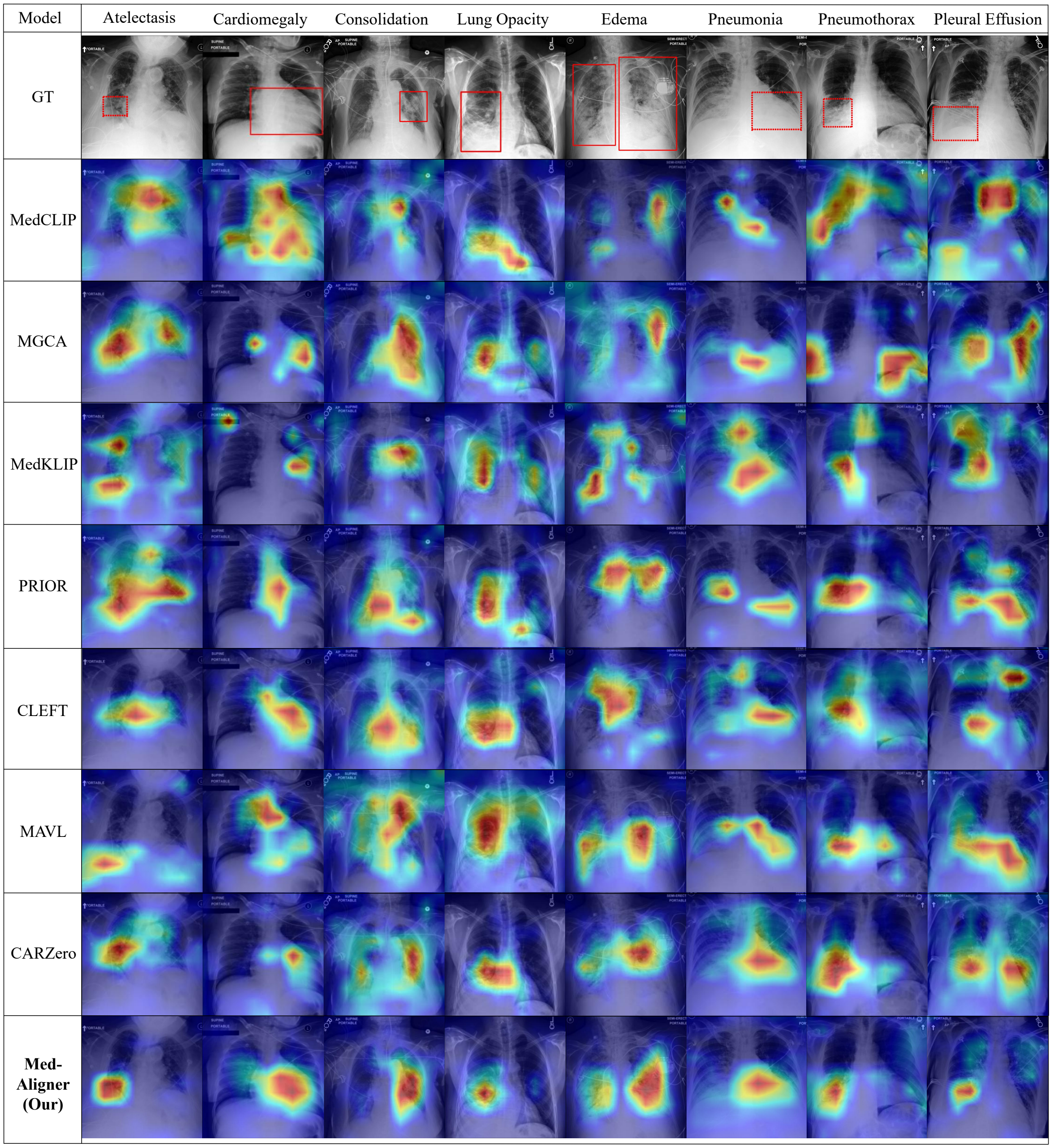

Fig. 7. Attention maps for eight disease categories are visualized on the MS-CXR dataset, comparing MedAligner with seven baseline methods. Red bounding boxes indicate the ground truth regions relevant to phrase grounding. Highlighted pixels correspond to higher activation weights, reflecting stronger associations between specific diagnostic terms and image regions.

local correspondence between diagnostic terms and lesion regions, dynamically extracting high-importance terms from clinical reports. It outperforms both MedKLIP and MAVL, demonstrating superior detection accuracy. The PTS further enhances precision and overall detection performance, highlighting the robustness of MedAligner in local lesion detection tasks.

Zero-Shot Classification. To assess the adaptability of the learned image encoder, we evaluated its performance on Zero-shot classification. A classification head was added to the image encoder and trained using cross-entropy loss, enabling the model to make predictions solely based on visual features.

Table V shows classification results on the MIMIC-5 $\times$ 200, COVID, RSNA Pneumonia, and NIH ChestX-ray datasets. MedAligner consistently outperformed all baseline methods, demonstrating strong generalization across diverse datasets. These results highlight MedAligner's ability to effectively capture transferable image representations, further validating its potential as a robust image-text pretraining model for medical applications.

TABLE IV
THE OBJECT DETECTION TASK IS COMPARED WITH STATE-OF-THE-ART METHODS ON THE RSNA PNEUMONIA DATASET, AND EVALUATION METRICS INCLUDING IOU (%), DICE (%) AND ACC (%) ARE REPORTED.

Model	IoU	Dice	Acc
MedCLIP	24.96	34.67	85.42
GLoRIA	33.30	46.17	89.31
BioViL	22.16	35.81	86.37
MGCA	35.52	48.65	86.70
MedKLIP	31.39	46.05	86.14
PRIOR	33.06	45.86	85.44
CLEFT	34.90	47.87	88.15
MAVL	35.44	51.02	88.84
CARZero	36.12	49.48	88.22
AFLoc	35.79	48.13	88.36
MedAligner (Our)	**37.92**	**51.28**	**89.63**

TABLE V
THE ACCURACY (%) OF THE ZERO-SHOT CLASSIFICATION TASK IS EVALUATED ON THE MIMIC-5 $\times$ 200, COVID, RSNA PNEUMONIA, AND NIH CHESTX-RAY DATASETS, AND COMPARED ACROSS STATE-OF-THE-ART METHODS.

Model	MIMIC-5 $\times$ 200	COVID	RSNA	NIH14
MedCLIP	52.50	77.70	79.90	58.84
GLoRIA	72.22	87.77	78.55	59.47
BioViL	73.38	80.10	78.46	61.33
MGCA	74.87	86.87	79.20	68.40
MedKLIP	51.94	83.26	74.65	79.40
PRIOR	76.83	86.27	74.73	80.84
CLEFT	75.47	84.18	78.25	78.95
MAVL	74.60	83.06	78.14	82.77
CARZero	76.06	86.80	78.55	83.16
AFLoc	76.20	87.80	73.52	83.87
MedAligner (Our)	**77.89**	**88.20**	**80.05**	**84.11**

Unpaired Image-Text Pre-training. To evaluate the local alignment performance of MedAligner on unpaired image-text data, we pre-trained the model using randomly shuffled unpaired image-text data from the MIMIC-CXR dataset. Specifically, we randomly shuffled all image-report pairs in the MIMIC-CXR dataset by 100%, simulating an extreme scenario where labels are mismatched or erroneous. Under these conditions, the model's effectiveness was validated through the phrase grounding task on the MS-CXR dataset. The results are shown in Table VI. The experimental outcomes demonstrate that the proposed local alignment strategy significantly improves performance on unpaired image-text data. MedAligner outperforms other local alignment-based methods, such as GLoRIA and MGCA.

Additionally, we conducted experiments using 50% randomly shuffled data pairs, simulating a more realistic scenario where some pairs are correctly matched while others contain errors. This setup allowed us to systematically assess the model's performance under various image-text pairing conditions. As shown in Table VI, after pretraining on the 50% shuffled data pairs, MedAligner still maintained stable performance on the MS-CXR dataset. By continuously optimizing the correspondence between text and image features, MedAligner remained robust even in the presence of incomplete reports.

C. Comparison with Large Multimodal Models

In this section, we compare MedAligner with two large multimodal models: LLaVA-Med and Qwen-VL. MedAligner is pretrained on medical data (chest X-rays and radiology reports), while LLaVA-Med and Qwen-VL are general-purpose models trained on large-scale multimodal datasets. The goal is to assess MedAligner's performance in medical image–text alignment relative to these models.

LLaVA-Med is a large-scale vision-language model that has been pretrained on extensive multimodal datasets, including medical images and their associated textual reports. It supports diverse medical tasks such as disease detection, image analysis, and clinical decision support by leveraging strong generalization and semantic understanding. **Qwen-VL** is another large multimodal model trained for complex vision–language tasks, excelling in image–text matching and cross-modal reasoning. In the medical domain, it has been applied to image description, report generation, and image–text alignment.

We evaluated the models on global alignment tasks (zero-shot classification) and local alignment tasks (object detection, phrase grounding). As shown in Table VIII, LLaVA-Med and Qwen-VL achieved higher accuracy than MedAligner on MIMIC-5 $\times$ 200 and COVID classification. However, in local alignment, MedAligner outperformed both models: it achieved higher IoU, Dice, and Acc on RSNA (Table IX) and higher CNR in phrase grounding on MS-CXR (Table X). These gains stem from MedAligner's ability to align diagnostic terms with specific image regions. By contrast, LLaVA-Med and Qwen-VL emphasize global semantics, limiting their performance in tasks requiring precise lesion localization.

D. Ablation Study

To better understand the performance of MedAligner, we conducted a series of ablation experiments focusing on the impact of different loss designs and the number of progressive training steps on local alignment on key tasks.

Impact of Loss Functions. We evaluated three loss configurations of MedAligner on the Phrase Grounding. The three loss configurations included: (1) global contrastive loss ($\mathcal{L}_g$), applied only to global image and text embeddings to establish a coarse-grained alignment; (2) global contrastive loss combined with a local loss using only one sparsification layer ($\mathcal{L}_g + \mathcal{L}_l'$); and (3) global contrastive loss combined with joint local loss ($\mathcal{L}_g + \mathcal{L}_l$), which uses two sequential sparsification layers to jointly pixel-word sparsification to achieve salient local alignment. Table VII summarizes the results. Models incorporating joint local loss ($\mathcal{L}_g + \mathcal{L}_l$) consistently outperformed those using only the global loss or the global loss with the local loss using one sparsification layer across all metrics. Specifically, this configuration achieves the highest CNR scores in the phrase grounding task, highlighting the effectiveness of local alignment enhanced by the two sequential sparsification layers.

Impact of Progressive Training Steps on Local Alignment. We evaluated the effect of progressive training on phrase grounding using the MS-CXR dataset, with MedAligner trained under $T = 1$ to 5 steps (Figure 8). With $T = 1$ and 2, the model struggled to establish precise correspondences

TABLE VI

EVALUATION OF MEDALIGNER TRAINED ON UNPAIRED IMAGE-TEXT DATA FOR PHRASE GROUNDING ON THE MS-CXR DATASET USING CONTRAST-TO-NOISE RATIO (CNR) SCORES ACROSS EIGHT DISEASE CATEGORIES.

Model	Atelectasis	Cardiomegaly	Consolidation	Lung Opacity	Edema	Pneumonia	Pneumothorax	Pleural Effusion
GLoRIA (100% unpaired)	0.3349	0.2794	0.3981	0.4177	0.1962	0.3127	0.2798	0.3136
MGCA (100% unpaired)	0.4391	0.2951	0.4095	0.4292	0.2102	0.5076	0.2924	0.3711
MedAligner (100% unpaired)	**0.5228**	**0.3923**	**0.5025**	**0.4956**	**0.4865**	**0.5523**	**0.4217**	**0.6591**
GLoRIA (50% unpaired)	0.4486	0.3050	0.4359	0.4479	0.2069	0.4941	0.3017	0.3533
MGCA (50% unpaired)	0.4620	0.2966	0.4363	0.4897	0.2352	0.6071	0.3569	0.4338
MedAligner (50% unpaired)	**0.6346**	**0.4149**	**0.5344**	**0.5431**	**0.5889**	**0.6836**	**0.6021**	**0.7298**

TABLE VII

PHRASE GROUNDING PERFORMANCE UNDER DIFFERENT LOSS CONFIGURATIONS IS EVALUATED ON THE MS-CXR DATASET, USING CONTRAST-TO-NOISE RATIO (CNR) SCORES ACROSS EIGHT DISEASE CATEGORIES.

	Atelectasis	Cardiomegaly	Consolidation	Lung Opacity	Edema	Pneumonia	Pneumothorax	Pleural Effusion
$\mathcal{L}_g$	0.70±0.04	0.57±0.03	0.76±0.03	0.55±0.04	0.68±0.04	0.69±0.11	0.60±0.07	0.76±0.06
$\mathcal{L}_g + \mathcal{L}_l'$	0.64±0.04	0.58±0.02	0.72±0.04	0.62±0.12	0.69±0.07	0.75±0.21	0.67±0.05	0.97±0.18
$\mathcal{L}_g + \mathcal{L}_l$	**0.83**±0.06	**0.61**±0.01	**0.81**±0.02	**0.72**±0.07	**0.73**±0.08	**0.85**±0.05	**0.72**±0.08	**1.02**±0.10

TABLE VIII

THE ACCURACY (%) OF THE ZERO-SHOT CLASSIFICATION TASK IS EVALUATED ON THE MIMIC-5 × 200, COVID, AND RSNA PNEUMONIA DATASETS AND COMPARED WITH LLAVA-MED AND QWEN-VL

Model	MIMIC-5 × 200	COVID	RSNA
LLaVA-Med	**79.76**	93.67	78.25
Qwen-VL	70.32	**94.60**	76.03
MedAligner (Our)	77.89	88.20	**80.05**

TABLE IX

THE OBJECT DETECTION TASK IS COMPARED WITH LLAVA-MED AND QWEN-VL ON THE RSNA PNEUMONIA DATASET, AND EVALUATION METRICS INCLUDING IOU (%), DICE (%) AND ACC (%) ARE REPORTED.

Model	IoU	Dice	Acc
LLaVA-Med	22.31	33.64	87.41
Qwen-VL	19.08	27.19	83.81
MedAligner (Our)	**37.92**	**51.28**	**89.63**

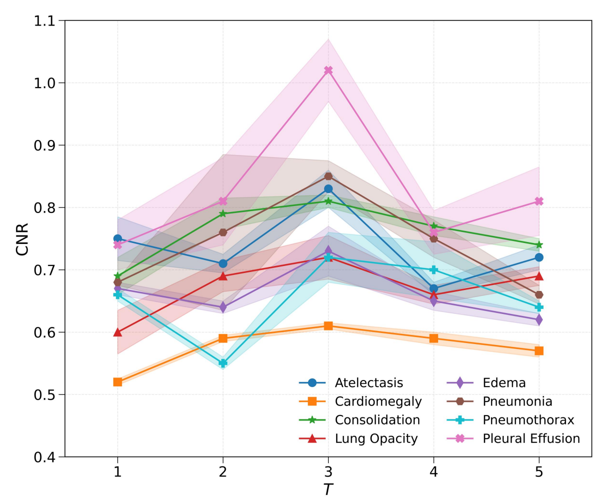

Fig. 8. The performance of MedAligner on the phrase grounding task is evaluated on the MS-CXR dataset, with Contrast-to-Noise Ratio (CNR) scores computed across eight disease categories under different numbers of progressive training steps T.

between lesion regions and textual descriptions, resulting in lower CNR scores. The performance improved significantly with $T = 3$, achieving the highest average CNR values across eight disease categories. However, when $T = 4$ and 5, scores declined slightly due to over-refinement, where essential terms (e.g., severity or location) were omitted. Based on these results, $T = 3$ was selected as the optimal progressive training step number for MedAligner, as it strikes a balance between retaining critical keywords and maintaining sufficient semantic coverage to enhance localization accuracy. Furthermore, when $T = 1$, MedAligner does not include the progressive training strategy. We compared the phrase grounding results of MedAligner at $T = 1$ and $T = 3$ (Figure 8). The results show that the progressive training strategy (i.e., $T = 3$) leads to a higher average CNR score. It also significantly improves the model's alignment accuracy and localization precision in the lesion regions compared to the case without progressive training (i.e., $T = 1$).

VI. CONCLUSION

We propose MedAligner, a framework for medical multi-modal alignment. It constructs a learnable word–pixel similarity matrix, refined by two sparsification layers to capture semantic links between diagnostic terms and salient image regions. MedAligner further samples key terms for local contrastive alignment and adopts a progressive training strategy that gradually shifts from global to local alignment by reconstructing reports paired with images. Experiments on multiple benchmarks show that MedAligner achieves state-of-the-art results in phrase grounding, image–text retrieval, and zero-shot classification. Despite its effectiveness, the training process is relatively complex, which may limit its applicability in resource-constrained settings. Future directions include knowledge distillation for efficiency and extending

TABLE X
THE PERFORMANCE OF THE PHRASE GROUNDING TASK IS COMPARED WITH LLAVA-MED AND QWEN-VL ON THE MS-CXR DATASET, USING CONTRAST-TO-NOISE RATIO (CNR) SCORES ACROSS EIGHT DISEASE CATEGORIES.

Model	Atelectasis	Cardiomegaly	Consolidation	Lung Opacity	Edema	Pneumonia	Pneumothorax	Pleural Effusion
LLaVA-Med	0.6385	0.4275	0.5655	0.5533	0.4425	0.5079	0.5443	0.7409
Qwen-VL	0.5665	0.3787	0.5305	0.5363	0.4708	0.2754	0.6519	0.5955
MedAligner (Ours)	**0.8312**	**0.6058**	**0.8144**	**0.7162**	**0.7335**	**0.8489**	**0.7249**	**1.0207**

MedAligner beyond chest X-rays to MRI, ultrasound, and broader multimodal tasks.

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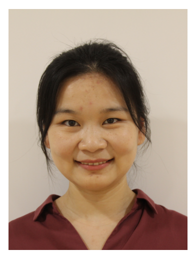

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